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Secure IoT-enabled edge computing ultrasonic device for structural health monitoring in inaccessible metallic structures

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Abstract

Structural health monitoring (SHM) in inaccessible or hazardous regions requires compact, autonomous, and secure devices capable of local signal processing and wireless data transmission. Conventional wired systems are difficult to deploy and often lack flexible connectivity and robust cybersecurity. This work presents an IoT-enabled edge computing device for ultrasonic SHM that integrates a piezoelectric actuator–sensor pair with a 32-bit ARM Cortex-M4 microcontroller for on-board signal conditioning and damage index (DI) computation, and an ESP32 microcontroller for secure wireless communication. The STM32 generates 200 kHz tone-burst excitation, applies fourth-order Butterworth high-pass and low-pass filters to captured ultrasonic responses, and computes a quantitative DI. The ESP32 implements AES-256 encryption, TLS-based MQTT (MQTTS), and an exponential backoff algorithm to publish encrypted telemetry to a HiveMQ cloud broker. A custom C#.NET desktop application subscribes to the broker, decrypts the data, and provides real-time visualization and local storage. Experimental validation on a mild steel plate ($570 \times 205 \times 10$ mm) with a $1 \times 70 \times 5$ mm artificial crack showed that the embedded DI closely matched offline calculations (2.05 vs. 2.01), confirming on-board processing accuracy. The device achieved encryption times of approximately $660 \mu\text{s}$ and TLS handshake times of about 1.49 s, while maintaining stable memory usage and low publish latency under MQTTS. Repeatability tests across five trials demonstrated clear separation between intact and damaged conditions with minimal variation in DI. The proposed device provides a compact, low-cost, and secure SHM solution suitable for inaccessible structural regions. By combining active ultrasonic sensing, edge computing, and encrypted IoT telemetry, the system enables reliable, real-time monitoring without transmitting raw ultrasonic waveforms.

1. Introduction

Structural health monitoring (SHM) is an interdisciplinary field devoted to continuous assessment of structural integrity using sensors, data analytics, and decision-support tools [1, 2]. Over the past few decades, SHM implementations have evolved from wired sensor networks to advanced wireless sensors and IoT technologies, enabling real-time monitoring of bridges, buildings, aerospace components, and other infrastructure [3–6]. Early SHM systems with tethered sensors were often costly and impractical to deploy broadly, whereas modern wireless-IoT architectures achieve cost-effective, scalable, and remote monitoring [7–9]. However, many SHM systems still face challenges when deployed in inaccessible or hard-to-reach structural regions, where wired sensing and manual inspection are impractical. This further motivates the need for compact, wireless, and autonomous devices. Early wireless SHM prototypes demonstrated that untethered ‘smart sensor’ nodes could perform local data processing and match the performance of wired systems. A wireless sensing unit equipped with a piezoelectric actuator and sensor was introduced to excite a structure and analyze its vibrational response on-board [10]. This

pioneering device identified structural damage by detecting changes in the response signals, proving that wireless nodes can embed intelligence to reduce data transmission while still detecting anomalies. This approach was further expanded by deploying a network of 14 high-resolution wireless accelerometers on a full-scale. Their system captured the bridge's modal frequencies and damping characteristics under traffic loads with accuracy comparable to a wired setup, validating the feasibility of wireless SHM in real infrastructure [4]. These results underscored the importance of on-node signal conditioning (e.g. local amplification and filtering) to improve measurement quality in wireless sensors. Subsequent advances in wireless SHM focused on improving sensor sensitivity and energy efficiency. One notable challenge is detecting very low-amplitude vibrations under ambient conditions. Zhu *et al* [11] addressed this by developing a high-sensitivity wireless accelerometer with a low-noise MEMS sensor and custom electronics, achieving a resolution on the order of $0.06 \mu\text{g}$. This level of sensitivity enables detection of subtle dynamic responses indicative of early-stage damage while maintaining low power consumption. Heo and Jeon [12] similarly demonstrated remote monitoring capabilities by deploying a Bluetooth-based sensing unit on a suspension bridge that transmitted real-time vibration data over a TCP/IP network. Their system successfully identified the bridge's natural frequencies and damping ratios, illustrating that even in 2009, wireless sensors could support real-time, remote structural diagnostics. Collectively, these innovations in wireless sensing laid the groundwork for IoT-based SHM, where distributed intelligent nodes continuously monitor and report structural health status [5]. In parallel with improvements in networking, SHM sensing techniques have advanced from passive vibration monitoring to active ultrasonic methods that deliberately inject diagnostic signals. Piezoelectric transducers (PZTs) play a central role in active SHM: they serve as actuators to send guided ultrasonic waves through a structure and as sensors to detect the waves' reflections and transmissions. Giurgiutiu [2] provides a foundational review of piezoelectric wafer active sensors, explaining how changes in vibrational responses or guided wave signals can indicate structural damage. Lynch's wireless unit mentioned above is an early example of active sensing in a wireless context, as it used a PZT actuator to excite ultrasonic waves in a test plate and performed on-board analysis to detect anomalies [10]. Compared to passive approaches, active ultrasonic sensing often offers greater sensitivity and can localize damage by examining how introduced waves are affected by structural flaws. Recent innovations have further enhanced active ultrasonic SHM devices. Choi *et al* [13] developed a laser-powered wireless ultrasonic system for aircraft monitoring that operates without batteries. Their device generated and received Lamb waves in aircraft panels and successfully detected damage (including small cracks and corrosion) with accuracy comparable to conventional wired systems. By harvesting energy from a laser beam, this system demonstrated the feasibility of powering sensors in hard-to-reach locations. Shrestha *et al* [14] introduced a programmable ultrasonic sensor platform with a 32-bit microcontroller, which allowed software control of excitation frequency and on-board digital signal processing. The device could transmit either raw ultrasonic waveforms or computed damage indices depending on the situation, thereby reducing data transmission when only summary damage metrics were needed. Similarly, Pertsch *et al* [15] developed a stand-alone ultrasonic unit that launched Rayleigh surface waves along a structure and autonomously detected surface crack propagation, showing the potential for long-term, self-contained damage monitoring. Guided wave techniques have proven especially effective for inspecting large structures. Staszewski *et al* [16] presented an overview of using guided ultrasonic waves (such as Lamb waves) for damage detection in aerospace structures, detailing how stress wave propagation is altered by structural discontinuities. Building on this foundation, Muller *et al* [17] employed the total focusing method to process Lamb wave data from a 16-transducer array, achieving high-resolution imaging of a ~ 10 mm defect in an aluminum plate. Their results showed that even sub-wavelength flaws could be detected and localized. Abbas and Shafiee [3] similarly found that guided waves are effective for detecting various damage types (e.g. cracks and corrosion) over long distances, but also noted the difficulty of interpreting complex wave signals to characterize the exact shape or orientation of a defect. This challenge has motivated the use of more sophisticated signal processing to analyze ultrasonic data. To extract more information from SHM signals, researchers have applied advanced signal processing methods. Amezcua-Sanchez and Adeli [1] reviewed techniques such as wavelet transforms, empirical mode decomposition, and Hilbert–Huang transforms for enhancing damage detection in noisy environments. These approaches help reveal subtle nonlinear patterns and transient features in vibration or ultrasonic data that traditional Fourier analysis might miss, thereby improving the identification of early or hidden damage. The incorporation of IoT technologies into SHM has fundamentally expanded the scope and capabilities of monitoring systems. IoT-based SHM architecture consists of distributed sensor nodes (edge devices) that communicate via wireless networks (e.g. Wi-Fi, cellular, or LPWAN) with cloud or server platforms for data aggregation and analysis [5]. Recent studies have demonstrated the application of edge computing in SHM systems to support local data processing, reduce

transmission demands, and improve real-time monitoring capability [18]. This connectivity enables real-time data access, remote visualization of structural conditions, and scalable deployment of large sensor networks. Scuro *et al* [5] also emphasized IoT's potential to support dense networks of smart sensors and real-time diagnostics, while also identifying persistent challenges such as time synchronization and cybersecurity in distributed environments. To improve synchronization, Lamonaca *et al* implemented a layered IoT architecture with precise time-stamping at the sensor, gateway, and server levels [19]. Their system achieved sub-second latency and accurate correlation of events (e.g. acoustic emissions) from multiple sensors, enhancing the reliability of real-time damage detection. Effective data communication is critical in IoT-driven SHM. Soni and Makwana [9] provided a detailed review of MQTT, highlighting that its lightweight publish/subscribe model (in which sensor nodes send data to a broker and client applications subscribe to relevant topics) incurs minimal network overhead and is well-suited for low-power, bandwidth-constrained SHM networks. Pierleoni *et al* [8] demonstrated a real-time building monitoring system using MQTT: a network of wireless sensors published acceleration and tilt measurements to a cloud-hosted broker, which relayed the data to visualization and analysis services with minimal latency. The system achieved continuous, low-latency monitoring and demonstrated MQTT's scalability and reliability in practice. However, despite MQTT's efficiency, concerns remain regarding secure data transmission and robust connectivity in real-world deployments. To address this, integrating encryption, secure transport protocols, and connection management strategies is essential. Low-latency monitoring and showcased MQTT's scalability and reliability in practice. Comprehensive IoT-based SHM frameworks have also been developed to integrate sensing, communication, and analytics. Mahmud *et al* [7] described a complete SHM platform with wireless sensor nodes, a central IoT gateway, and a cloud interface for data processing and storage. Their implementation included automated threshold-based damage alerts and power management strategies for the sensors, illustrating how SHM can function as an integrated IoT service. For high-volume data scenarios, Malik *et al* [20] proposed an industrial IoT architecture that distributes computation across edge, fog, and cloud layers. In their design, edge devices performed preliminary signal filtering and feature extraction, fog-level processors handled real-time diagnostics, and cloud servers managed long-term data storage and advanced analytics (including machine learning for damage classification). This distributed approach, supported by a high-throughput database for streaming data, reduced bandwidth usage and latency while enabling near-autonomous damage detection with minimal human intervention. As SHM systems scale up, managing the volume of sensor data and conserving energy have become crucial. Edge computing addresses these issues by processing data locally at the sensor node or gateway, significantly reducing communication overhead and latency. Studies have shown that edge processing can reduce SHM network traffic by an order of magnitude while preserving signal reconstruction quality and modal-frequency tracking performance [21]. Buckley *et al* [22] demonstrated a low-power edge SHM approach in which each wireless sensor node computed key features (e.g. dominant frequency, damping ratio) from its vibration measurements on-board and transmitted only these features periodically. This strategy drastically cut down data traffic and allowed the use of low-bandwidth networks (like LoRa), enabling long-term battery operation without sacrificing damage detection capability. Similarly, Bisio *et al* [23] developed an edge sensing platform using Narrowband IoT communication. Their sensor nodes performed local preprocessing and compression of the data so that only small packets were sent over the cellular network, and they incorporated energy-efficient design (including the potential for energy harvesting) to achieve multi-year continuous monitoring. Recent work has demonstrated TinyML implementation for guided-wave-based damage localization, enabling on-board inference in SHM using resource-constrained embedded devices. These include lightweight machine-learning and neural-network models deployed directly on embedded microcontroller platforms for SHM damage classification. By integrating such edge intelligence, SHM systems can provide immediate anomaly detection and further reduce dependence on constant cloud connectivity. Overall, the synergy of smart sensors, active ultrasonic techniques, IoT connectivity, and edge computing is driving the development of next-generation SHM systems that are autonomous, scalable, and capable of real-time predictive diagnostics. This evolution promises to enhance infrastructure safety and resilience by enabling continuous, proactive monitoring and timely maintenance interventions. Recent studies have demonstrated the feasibility of edge-based ultrasonic SHM systems capable of real-time sensing and *in-situ* data processing, significantly reducing reliance on centralized computation [24, 25].

In this paper, we present a prototype IoT-enabled edge computing device for ultrasonic SHM that embodies these advances. The proposed system employs a PZT as both an actuator and a sensor, combined with two microcontrollers for local processing and connectivity. A 32-bit STM32 ARM Cortex-M4 microcontroller (with a floating-point unit) is used to generate and acquire ultrasonic signals and to perform on-board signal processing, including digital filtering and damage index (DI) calculation. A custom

PZT actuator circuit drives ultrasonic waves into the structure using an approximately 200 kHz tone-burst excitation, while a PZT sensor circuit captures the propagating waves.

The STM32 digitizes the sensor feedback and applies a high-pass filter (HPF) (cutoff ~ 50 kHz) and a low-pass filter (LPF) (cutoff ~ 250 kHz) to isolate the relevant frequency band, followed by computation of a quantitative DI representing the structural condition. The second microcontroller, an ESP32, manages IoT connectivity by establishing a Wi-Fi link and publishing the computed DI and related data to a remote server using the MQTT protocol. Communication between the STM32 and ESP32 is achieved via a UART (USART) interface.

To ensure secure and reliable data transmission, the ESP32 implements AES-256 encryption and MQTT over Transport Layer Security (MQTTS), along with an exponential backoff algorithm for robust communication under network variability. For validation, encrypted telemetry was successfully transmitted to a HiveMQ cloud broker, confirming reliable end-to-end communication. A custom C#.NET desktop application was developed to subscribe to the MQTT topic, decrypt incoming data using AES-256 with PKCS7 padding, and visualize the DI in real time, while also storing the data locally for post-analysis.

The system was experimentally validated on a mild steel specimen ($570 \text{ mm} \times 205 \text{ mm} \times 10 \text{ mm}$) containing an artificial crack ($1 \text{ mm} \times 70 \text{ mm} \times 5 \text{ mm}$). The results demonstrate that the proposed device can effectively detect structural damage, highlighting the feasibility of integrating ultrasonic sensing, embedded processing, and secure IoT communication in a compact SHM platform.

Despite significant progress in IoT-based SHM systems, many existing approaches rely on transmitting raw ultrasonic data to cloud platforms, leading to increased bandwidth consumption, higher latency, and greater exposure to security risks. Additionally, the integration of real-time on-board signal processing with secure communication within a compact embedded platform remains limited.

To address these challenges, the proposed system performs all signal processing locally on the STM32 microcontroller, and transmits only the computed DI via the ESP32. This approach significantly reduces data transmission requirements, minimizes latency, and enhances security by limiting the amount of transmitted data. The desktop application is used solely for visualization and data storage, without participating in the signal processing pipeline, thereby ensuring that all critical computation occurs at the edge.

This work demonstrates the system-level integration of real-time ultrasonic signal processing, edge computing, and secure communication within a unified embedded SHM platform. The focus is on achieving efficient, secure, and real-time structural monitoring through integrated system design, rather than developing a new signal processing algorithm. The main contributions of this work are summarized as follows:

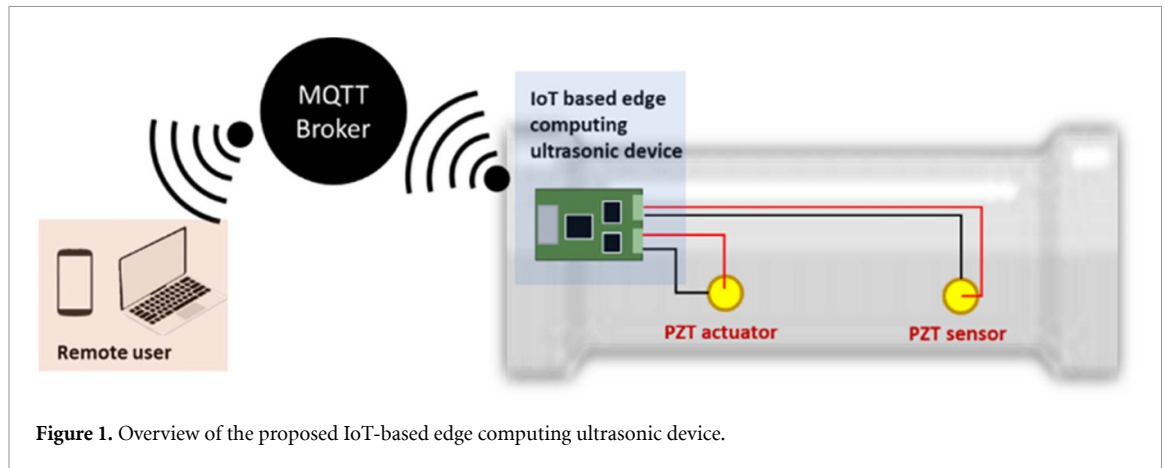
- Development of a compact dual-microcontroller-based ultrasonic SHM system integrating STM32 for real-time signal processing and ESP32 for secure wireless communication.
- Implementation of edge computing by performing on-board signal processing and transmitting only processed damage indices instead of raw ultrasonic data.
- Integration of secure communication protocols, including AES-256 encryption and MQTT over TLS (MQTTS), within an embedded SHM system.
- Experimental validation of the proposed system on metallic structures, demonstrating real-time damage detection capability.
- Demonstration of a low-power, scalable architecture suitable for deployment in inaccessible or remote structural environments.

2. Materials and methods

2.1. Overview of the SHM device

The proposed IoT-based ultrasonic device for SHM integrates hardware, firmware, and software components to enable efficient damage detection and secure remote monitoring of metallic structures. By combining edge computing, ultrasonic sensing, and IoT communication, the system facilitates real-time monitoring in inaccessible or hazardous environments.

Figure 1 illustrates the overall system architecture, highlighting the interaction and roles of the key components. The system consists of the following core modules:



- **PZT actuator circuit:** Generates a 200 kHz tone-burst ultrasonic signal to excite the metallic structure.
- **PZT sensor circuit:** Captures ultrasonic waves reflected from the structure for analysis.
- **STM32 microcontroller:** Performs real-time signal acquisition, filtering, and computation of the DI.
- **ESP32 microcontroller:** Manages wireless communication and implements secure data transmission using AES-256 encryption and MQTTS, along with an exponential backoff mechanism for reliability.
- **Signal conditioning circuits:** High-pass and LPFs are used to remove noise and isolate the relevant frequency components.
- **Cloud broker (HiveMQ):** Receives encrypted telemetry data for remote monitoring and storage.
- **Desktop visualization application:** A C#.NET-based interface subscribes to the broker, decrypts the received data, visualizes real-time DI trends, and stores the data for post-analysis.
- **Remote server/user interface:** Provides access for monitoring and decision-making based on transmitted data.

The developed hardware prototype of the proposed system is presented in figure 2. The system adopts a dual-microcontroller architecture, where the STM32F407 is responsible for real-time ultrasonic signal processing and DI computation, while the ESP32 handles secure wireless communication using MQTTS. The analog front-end incorporates multiplexers and operational amplifiers to enable signal conditioning and multi-channel acquisition.

In the proposed architecture, all signal processing operations, including filtering and DI computation, are performed locally on the STM32 microcontroller. Only the computed DI is transmitted via the ESP32 using MQTTS, significantly reducing bandwidth requirements compared to systems that transmit raw ultrasonic waveforms. The desktop application is used solely for visualization and data storage and does not participate in signal processing, ensuring true edge computing with real-time decision-making at the embedded device level.

The block diagram in figure 3 outlines the high-level operational workflow. The PZT actuator excites the structure with a tone-burst ultrasonic signal, and the reflected signals are captured by the PZT sensor. These signals are processed by the STM32 microcontroller to compute the DI, which is then securely transmitted by the ESP32 to the MQTT broker. The remote interface visualizes and stores the received data for analysis and decision-making. This integrated architecture enables efficient and reliable SHM while minimizing data transmission overhead.

The following sections describe the actuator and sensor circuits, followed by details on firmware development, cloud communication and visualization, and signal processing for the DI.

2.2. Actuator circuit

The actuator circuit uses a non-inverting amplifier, coupling capacitor, programmable amplifier, and push-pull amplifier to generate a tone-burst signal that actuates the PZT. A 200 kHz tone-burst ultrasonic signal is generated to excite the metallic structure.

The excitation frequency of 200 kHz was selected to balance penetration depth and sensitivity. Lower frequencies provide deeper penetration but reduced sensitivity to small defects, while higher frequencies improve sensitivity but increase signal attenuation. The chosen frequency lies within a commonly used

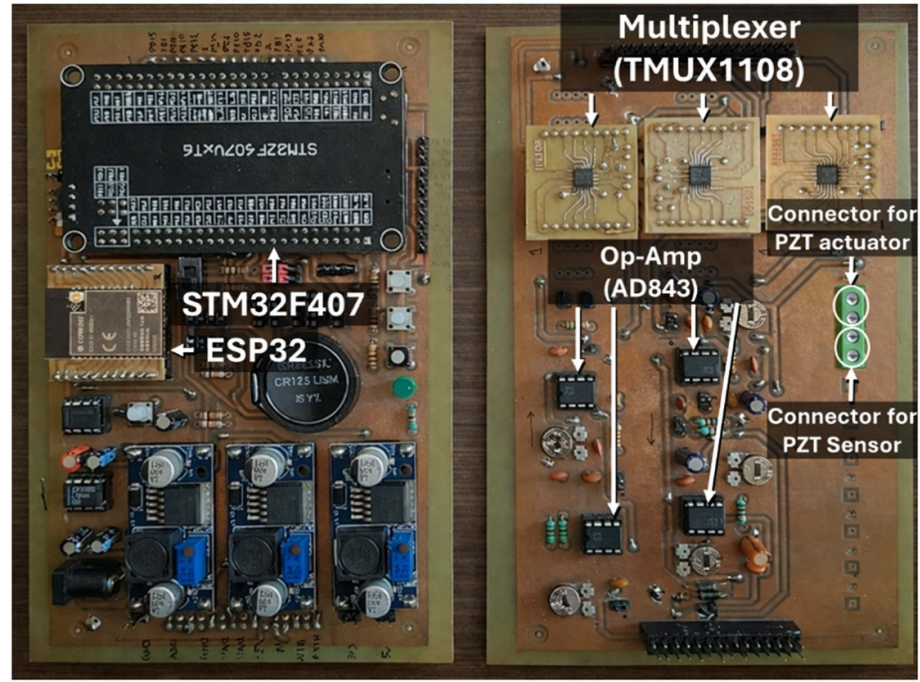


Figure 2. Hardware prototype of the proposed edge computing ultrasonic SHM system with STM32-based signal processing, ESP32-based secure communication, and analog front-end with PZT interfaces.

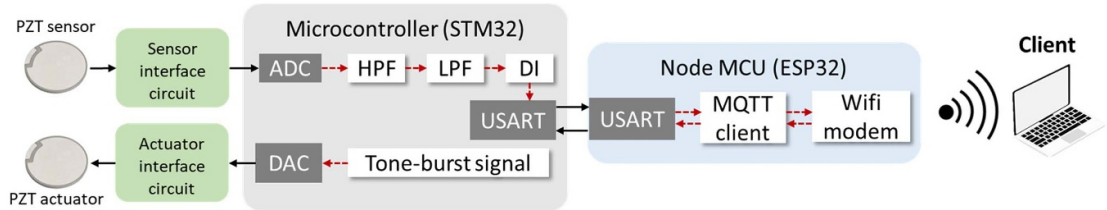


Figure 3. Block diagram of IoT-based edge computing ultrasonic device.

range for guided wave-based SHM applications and provides a practical compromise for detecting structural anomalies in metallic materials [26, 27].

The STM32 microcontroller's firmware generates a 6-cycle tone-burst signal from its DAC pin, represented by:

$$\text{tone}_{\text{burst}} = \sin 7t - \sin 6t \text{ for } 0 \leq t \leq \pi. \quad (1)$$

In order to achieve a maximum frequency of 200 kHz, data is transferred from memory to the DAC pin using the DMA controller. However, the resulting output voltage range of 0–1.28 V is insufficient to drive the PZT actuator and the dc-component of the tone-burst signal adds extra noise to the ultrasonic wave in the sensor circuit. Directly implementing a PZT driving amplifier at the microcontroller's output could distort the tone-burst signal. To address this issue, the actuator circuit was designed with two-stage amplifiers that include a coupling capacitor and push-pull amplifier, as shown in figure 4.

The first stage amplifier utilizes a non-inverting amplifier with a gain of 1.2. After that, a coupling capacitor removes any DC components from the tone-burst signal. The second stage addresses the issue of signal distortion caused by the maximum peak-to-peak voltage applied to the PZT, which occurs when the device is near structures. To solve this problem, a programmable amplifier was designed with a TMUX1108 multiplexer and four resistors at its feedback path, enabling the gain to be set to 6, 8, 10, or 12. The microcontroller controls the selection lines of the multiplexer. Additionally, a capacitor was added at the feedback path and a resistor at the amplifier output to compensate for the PZT capacitance. Finally, a push-pull amplifier was included in the circuit to increase the current for driving the PZT.

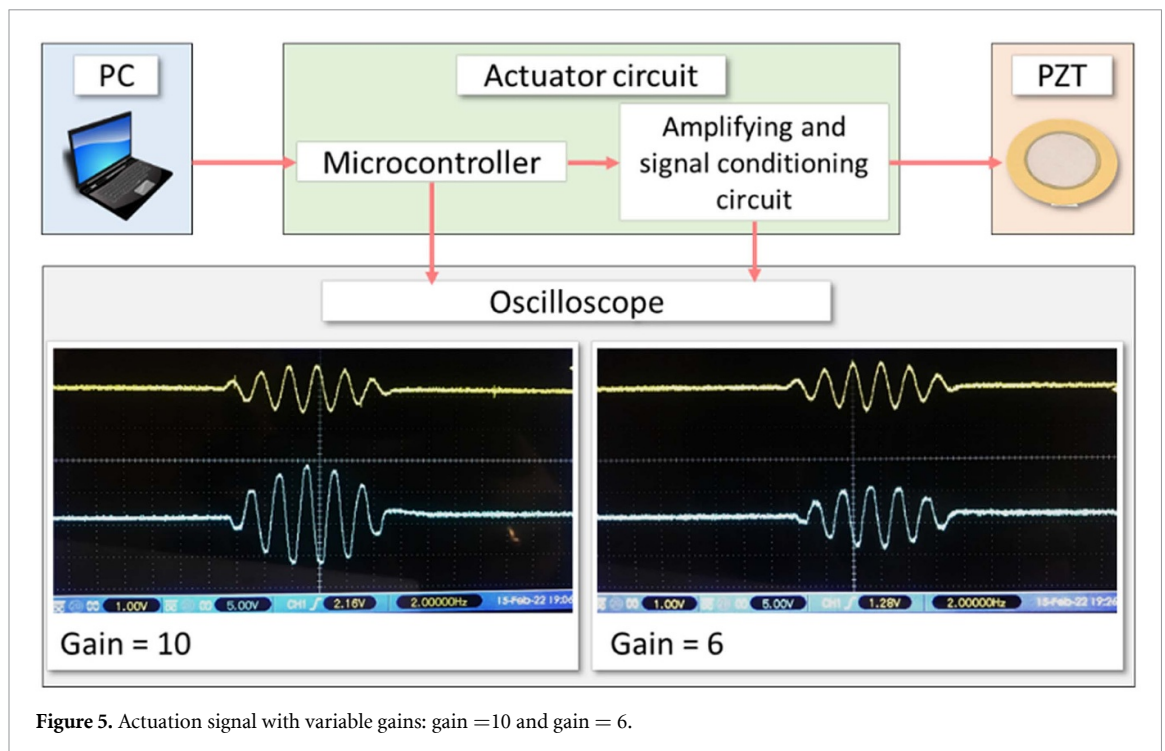
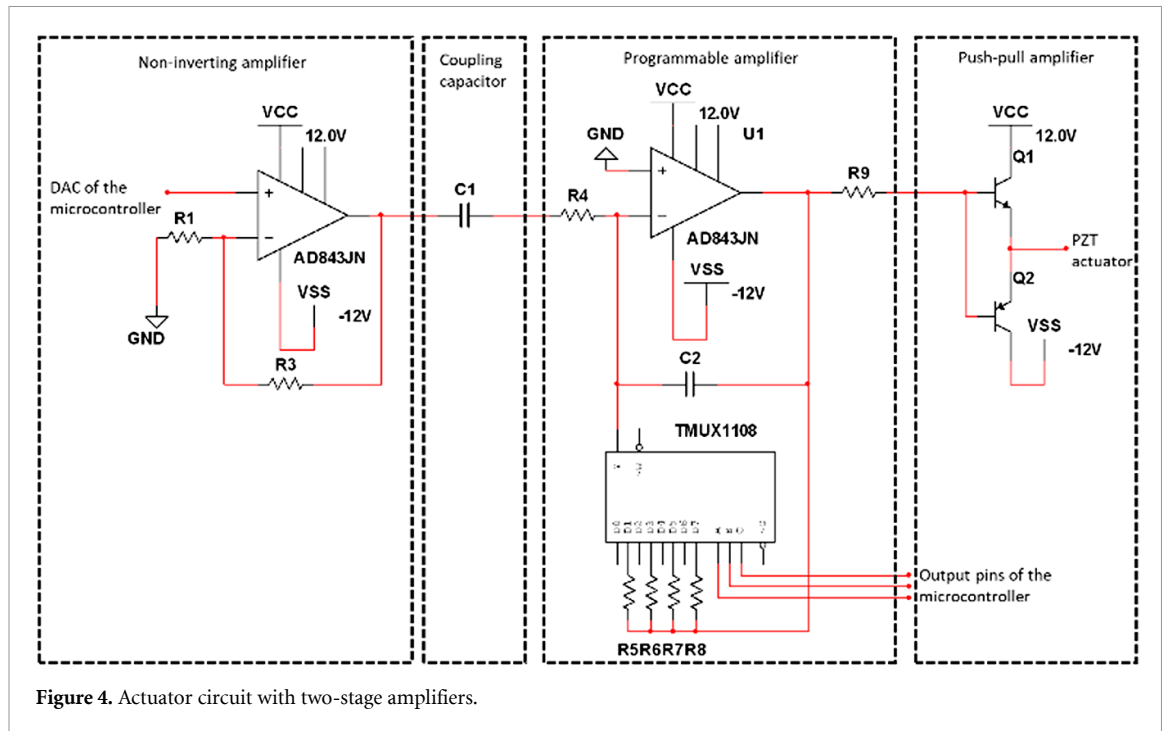


Figure 5 shows the experimental setup to test the actuation signal with a signal frequency of 200 kHz and its output on the oscilloscope. A host interface was used to configure the gain settings during testing; however, this interface is not required during normal operation, where all processing is performed on the embedded system. Figure 5 also shows the actuation signal with the gain of 10 and 6 using the designed programmable amplifier and hence can be used with different gain of the amplifier depending on the requirement with the structure to monitor and the placement of the sensors.

2.3. Sensor circuit

The sensor circuit (figure 6) is designed to gather ultrasonic waves from the PZT sensor. To accomplish this, an AD843 op-amp is used, which has a total power bandwidth of 3.9 MHz, an input resistance of

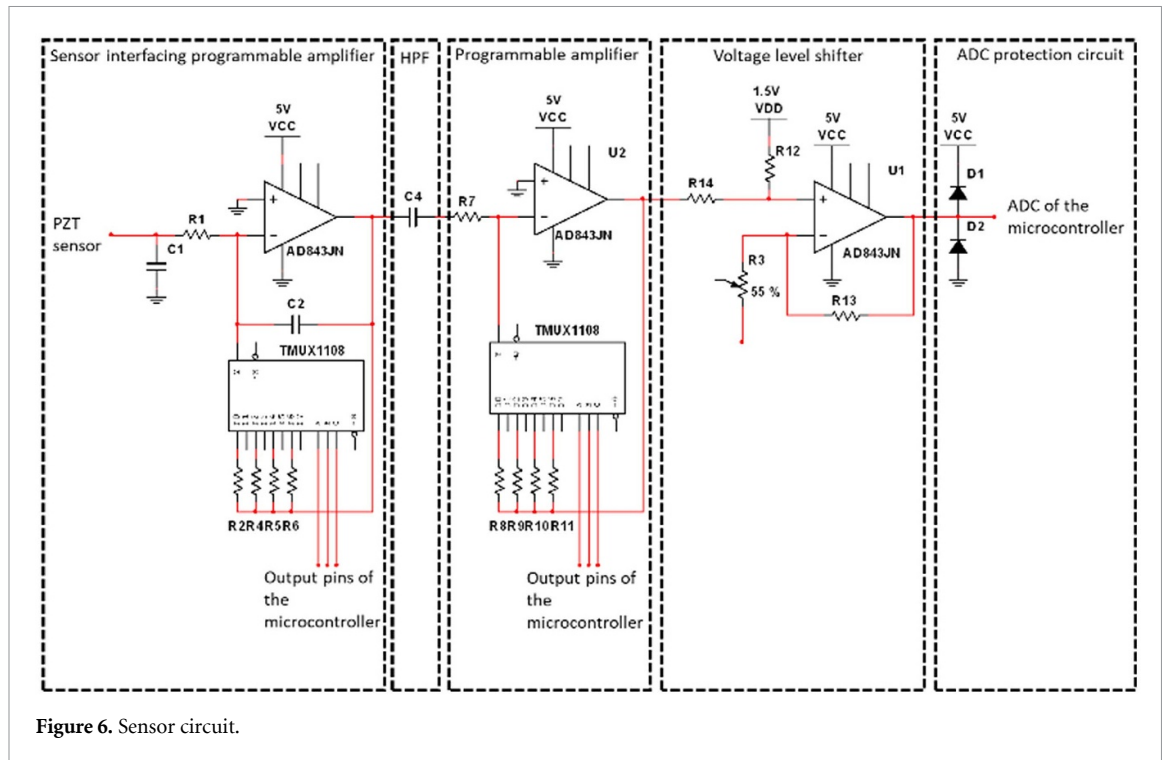


Figure 6. Sensor circuit.

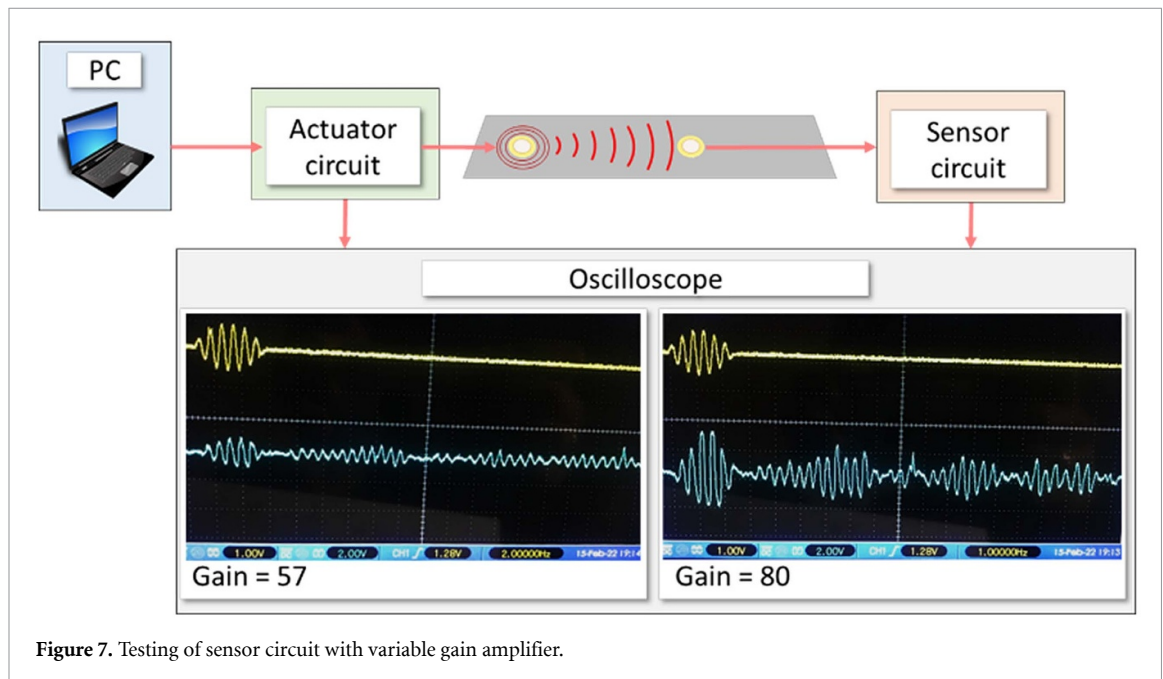


Figure 7. Testing of sensor circuit with variable gain amplifier.

10 G, and an input capacitance of 6 pF. A 12.2 pF capacitor was added to the circuit to match the PZT sensor's capacitance, thereby increasing the input capacitance to 18.2 pF. To handle amplitude variation resulting from structure thickness and sensor placement, a programmable amplifier with selectable gain was also included in the circuit.

To eliminate frequencies below 10 Hz, a high pass filter with a 10 Hz cut-off frequency was added. An additional programmable amplifier with adjustable gains was also included in the circuit for larger structures. The selectable gains for this amplifier are 1, 12, 24, 35, 46, 57, 69, and 80.

To digitize both positive and negative voltage, a voltage level shifter circuit was added using a voltage adder with the ultrasonic wave and a constant 1.5 V input. An ADC protection circuit was also included to limit the voltage level at the ADC pin to a range of 0–3.3 V and prevent damage.

To test the functionality of the sensor circuit, a command was sent from the PC to actuate the actuator circuit, as depicted in figure 7. The ultrasonic wave was captured and processed by the sensor circuit

and monitored via an oscilloscope to validate the circuit's performance. As demonstrated in figure 7, the actuation signal from the actuator circuit and the received ultrasonic wave from the sensor circuit were clearly visible, affirming the circuit's ability to detect ultrasonic waves from the PZT sensor. Additionally, to test the programmable amplifier's functionality, the ultrasonic wave was captured twice with two different gains selected by the multiplexer's selection line. The discrepancy in the received signal's amplitude, as illustrated in figure 7, confirmed that the designed programmable amplifier was proficient in collecting ultrasonic waves from the sensor.

2.4. Development of firmware

The device integrates two microcontrollers, ESP32 and STM32, each running FreeRTOS to enable efficient multitasking and synchronization for real-time operations.

1. ESP32 microcontroller:

- The ESP32 microcontroller manages communication and data transmission. two freeRTOS tasks are implemented:
 - **Task 1:** Initializes Wi-Fi communication, retrieves the current date and time using the Simple Network Time Protocol, and ensures a stable Wi-Fi connection.
 - **Task 2:** Receives the computed DI from the STM32 microcontroller via UART communication and publishes it securely to a remote server using the MQTT protocol.

2. STM32 microcontroller:

- The STM32 microcontroller is responsible for data acquisition and processing. Two FreeRTOS tasks are implemented:
 - **Task 1:** Generates a tone-burst signal to actuate the actuator circuit, captures the ultrasonic wave reflected from the sensor circuit, processes the signal through a HPF and a LPE, and computes the DI.
 - **Task 2:** Maintains a robust connection with the ESP32 microcontroller to ensure seamless data exchange.

The use of FreeRTOS in this dual-microcontroller configuration ensures precise task management, enabling reliable data acquisition, processing, and secure transmission in real-time. This architecture enhances the system's efficiency and robustness, critical for IoT-based applications. Figure 8 illustrates the process by which raw sensor data is collected, processed, and systematically organized within the STM32 microcontroller to enable efficient retrieval and analysis.

2.5. Cloud communication and real-time data visualization

The ESP32 microcontroller published encrypted telemetry packets using the MQTTS protocol to a HiveMQ cloud broker, ensuring data confidentiality and integrity through AES-256 encryption with PKCS7 padding. Each published message contained the device identification, timestamp, and computed DI. The HiveMQ web dashboard verified successful transmission, displaying both encrypted and decrypted payloads in real time.

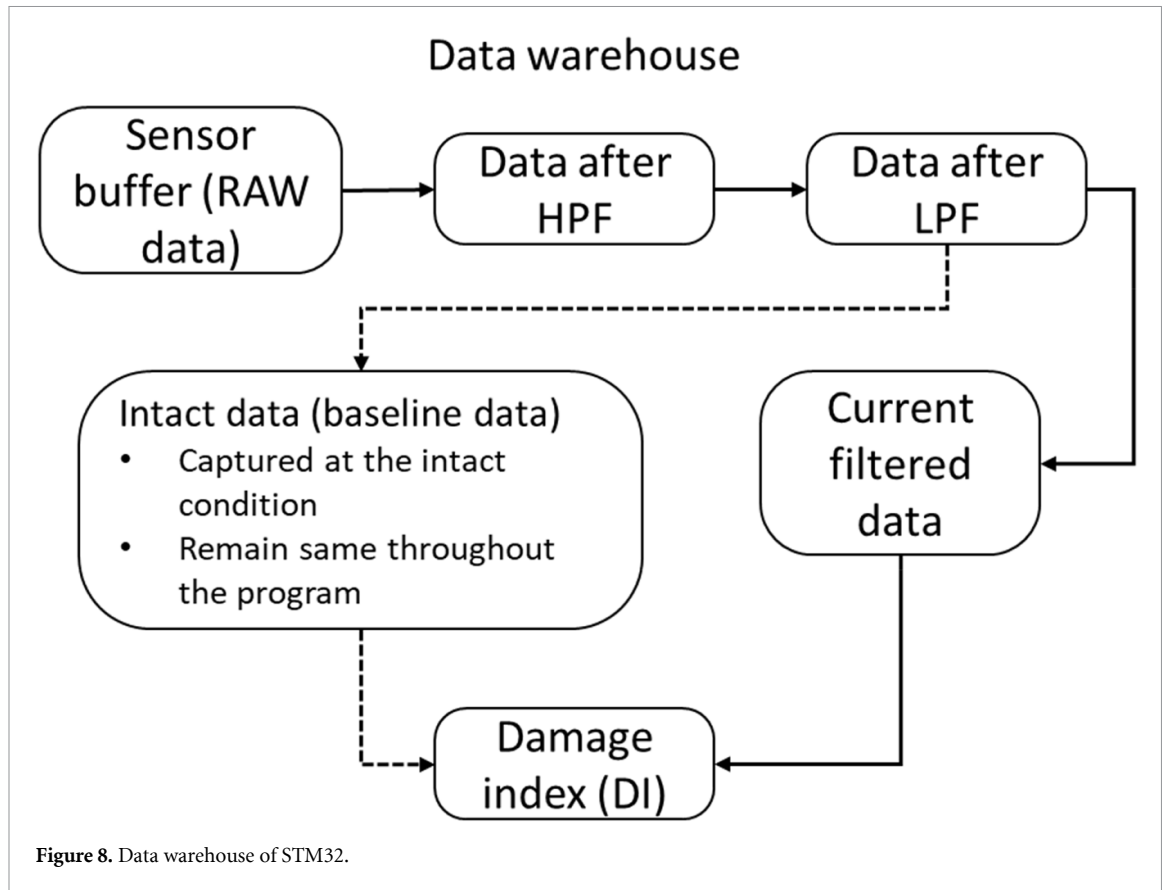
In parallel, a C#.NET desktop application was developed to provide a secure, local monitoring interface. The application used an MQTT client to subscribe to the same topic (IUD_test/DI), retrieve the encrypted telemetry, decrypt it using the identical AES-256 key and initialization vector, and dynamically plot the DI values. A dedicated 'DataFolder' directory was implemented to store decrypted messages for post-analysis and recordkeeping.

This dual-layer visualization, cloud-based and desktop-based, demonstrated the system's ability to securely transmit, decrypt, and visualize real-time SHM data, validating the reliability of the implemented IoT communication pipeline.

2.6. Signal processing for DI

The DI represents the change in the ultrasonic wave received from the structure. To calculate it, the captured ultrasonic wave is first filtered to remove any unwanted noise.

The device captures the ultrasonic wave at the sampling rate of 2.5 MSPS and actuates the PZT actuator with a frequency of 200 kHz. Therefore, the high pass filter with a cut-off frequency of 50 kHz and the low pass filter with a cut-off frequency of 250 kHz was designed to remove noise from the signal. The designed filters are fourth-order Butterworth IIR filters. The transfer function of the designed filter

**Table 1.** Filter coefficients for 50 kHz HPF.

Numerator				
b_0	b_1	b_2	b_3	b_4
0.848 475 295 52 44	−3.393 901 182 097	5.090 851 773 146	−3.393 901 182 097	0.848 475 295 52 44
Denominator				
a_0	a_1	a_2	a_3	a_4
1	−3.671 729 089 162	5.067 998 386 734	−3.115 966 925 202	0.719 910 327 29 19

is shown in equation (2).

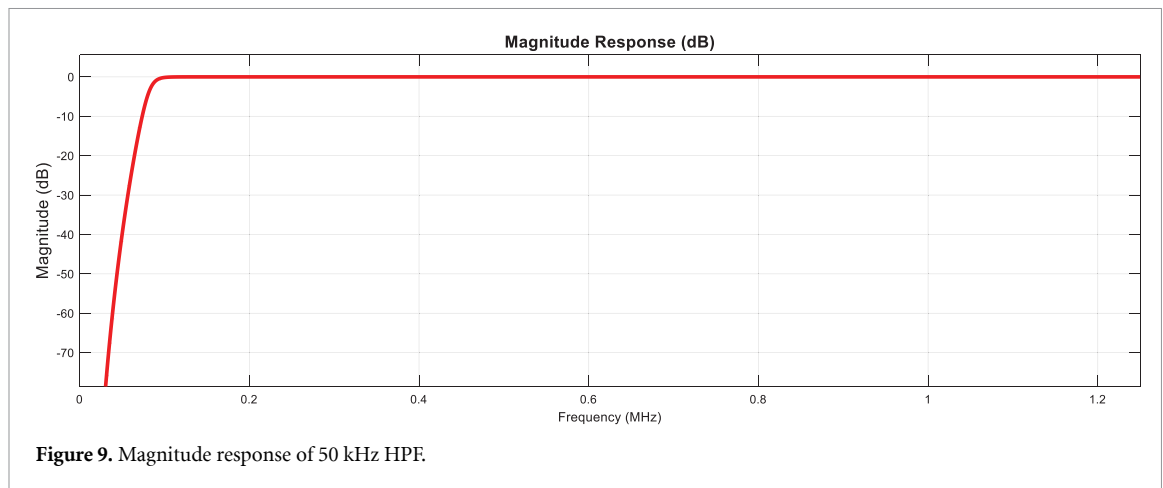
$$H(z) = \frac{b_0 + b_1z^{-1} + b_2z^{-2} + b_3z^{-3} + b_4z^{-4}}{a_0 + a_1z^{-1} + a_2z^{-2} + a_3z^{-3} + a_4z^{-4}}. \quad (2)$$

The filter was realized using the direct form II structure. The calculated filter coefficients and magnitude response of the designed HPF with 50 kHz cutoff frequency are shown in table 1 and figure 9, respectively. Similarly, the calculated filter coefficients and magnitude response of the designed LPF with 250 kHz cutoff frequency are shown in table 2 and figure 10, respectively.

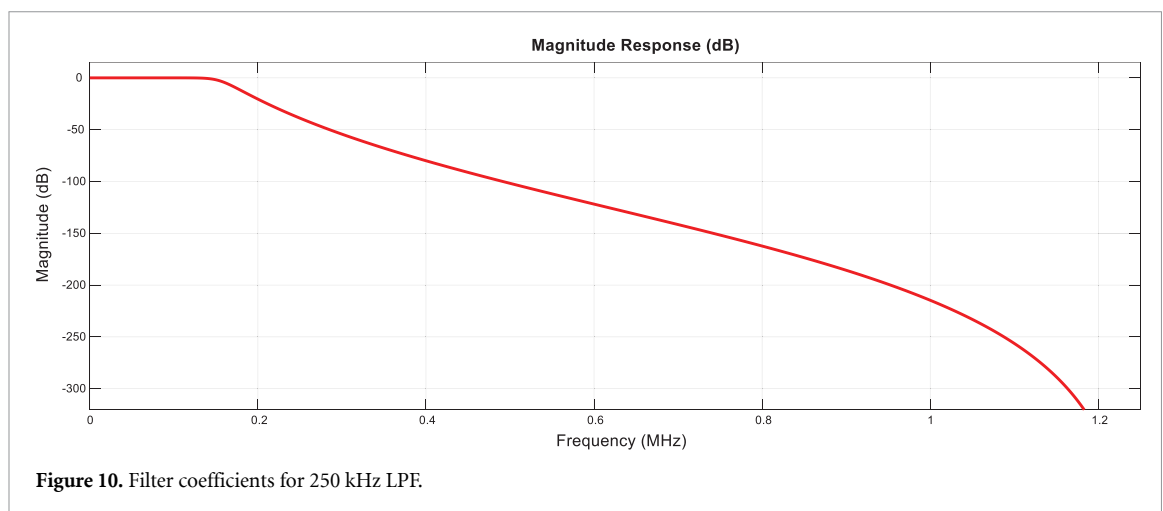
After filtering the wave, the DI was calculated using the equation:

$$DI_{\text{TDRMS}} = \frac{\sum_{i=1}^n |S_m(i) - S_b(i)|^2}{\sum_{i=1}^n |S_b(i)|^2} \quad (3)$$

where, S_m is a digitized ultrasonic wave and S_b is the digitized baseline wave from intact condition.

**Table 2.** Filter coefficients for 250 kHz LPF.

Numerator				
b_0	b_1	b_2	b_3	b_4
0.004 824 343 357 716	0.019 297 373 430 86	0.028 946 060 14 63	0.019 297 373 430 86	0.004 824 343 357 716
Denominator				
a_0	a_1	a_2	a_3	a_4
1	-2.369 513 007 182	2.313 988 414 416	-1.054 665 405 879	0.187 379 492 36 82



3. Result

The hardware prototype was validated through repeated experimental trials under identical conditions, confirming stable signal acquisition and consistent DI computation. The analog front-end demonstrated reliable amplification and noise suppression across different gain settings, as verified through oscilloscope measurements. These results confirm the robustness of the hardware design for real-time SHM applications.

3.1. Ultrasonic signal acquisition and on-board filtering

The performance of the edge device in extracting and conditioning ultrasonic responses was first evaluated using an intact mild steel specimen (560 mm × 250 mm × 10 mm). The device recorded (i) the raw digitized ultrasonic wave, (ii) the wave after the 50 kHz HPF, and (iii) the wave after the 250 kHz LPF. The experimental setup is shown in figure 11.

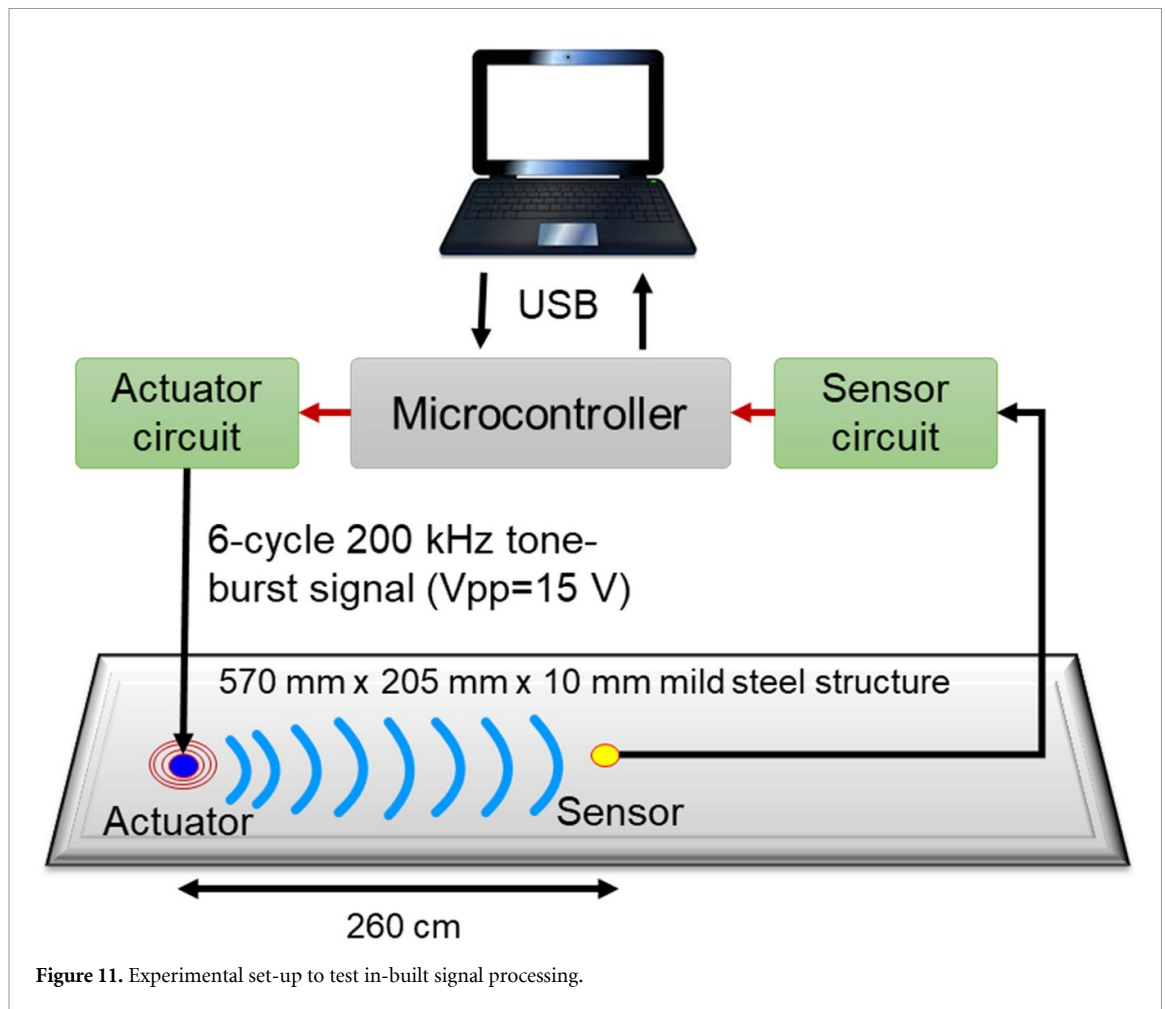


Figure 11. Experimental set-up to test in-built signal processing.

As illustrated in figure 12, the filtering stages effectively suppress low-frequency and high-frequency noise components while preserving the 200 kHz tone-burst content generated by the actuator. The FFT plots further verify that the fourth-order Butterworth HPF and LPF isolate the desired frequency band for reliable DI computation. These results confirm that the on-board digital filters perform as intended for real-time embedded signal conditioning.

3.2. Damage detection through embedded DI computation

To evaluate crack detection capability, a $1\text{ mm} \times 70\text{ mm} \times 5\text{ mm}$ artificial defect was introduced into the same specimen (figure 13). Ultrasonic responses were collected for both intact and damaged states.

Figure 14 shows a clear change in the waveform amplitude and shape when the crack is present, indicating a significant perturbation in wave propagation. The DI calculated offline on a PC for the damaged condition was 2.01, while the embedded system independently computed a DI of 2.05. The close agreement between embedded and offline calculations demonstrates that on-board processing is sufficiently accurate to detect structural defects without external computation. The present study evaluates a defect of size $1 \times 70 \times 5\text{ mm}$, which represents a relatively large crack. Previous studies have demonstrated the detection of smaller defects (e.g. on the order of $\sim 10\text{ mm}$) using guided wave techniques [17]. The sensitivity of the proposed system depends on factors such as excitation frequency, sensor placement, and signal processing methods. Future work will focus on detecting smaller defects and quantifying the minimum detectable crack size.

3.3. Secure telemetry transmission and real-time visualization

After confirming that the embedded device can compute a DI that distinguishes intact and damaged conditions, end-to-end secure telemetry was evaluated using the complete IoT setup shown in figure 15, which includes the STM32 ultrasonic sensing node, ESP32 communication module, local Wi-Fi access point, and cloud MQTT broker.

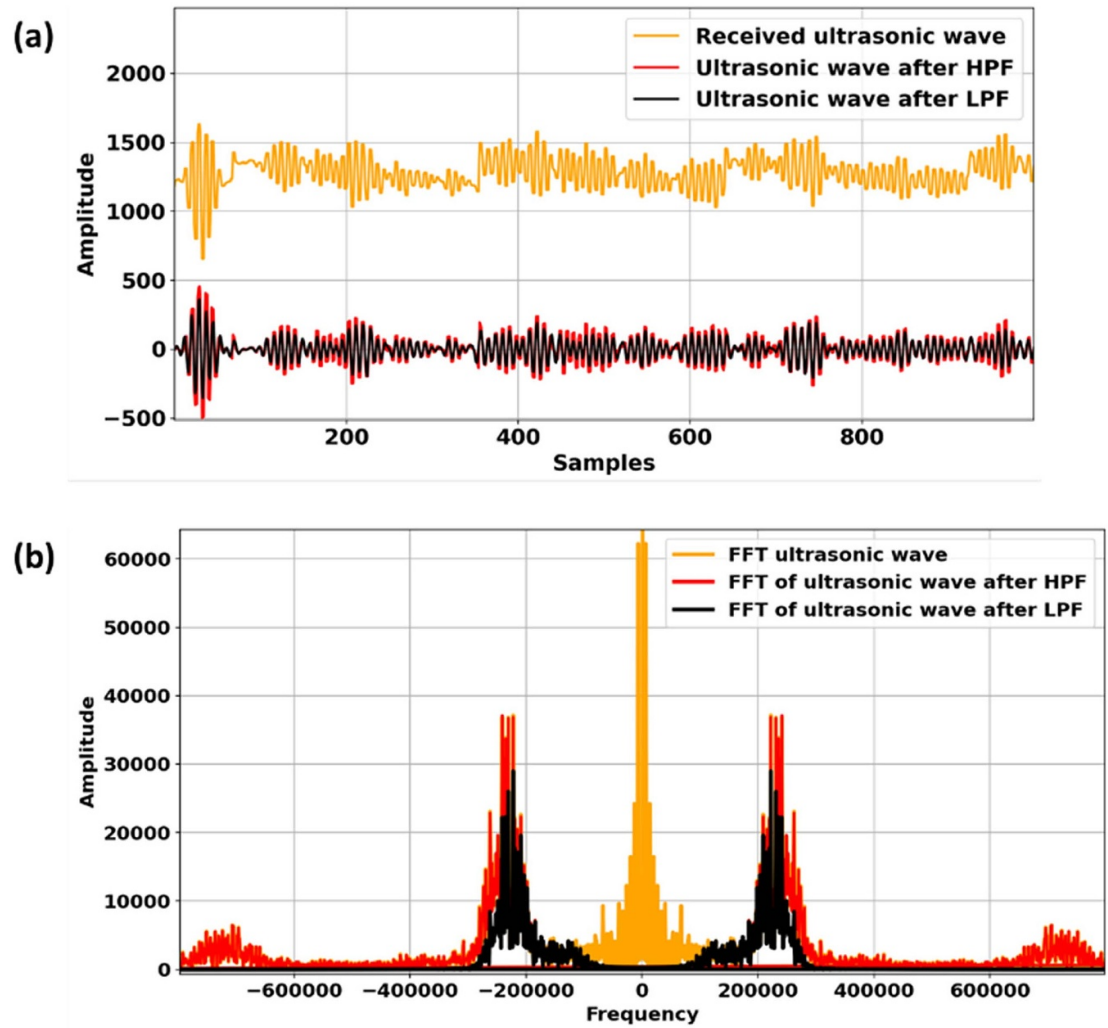


Figure 12. Digitized ultrasonic waves received from experimental set-up, (b) FFT of the received ultrasonic waves.

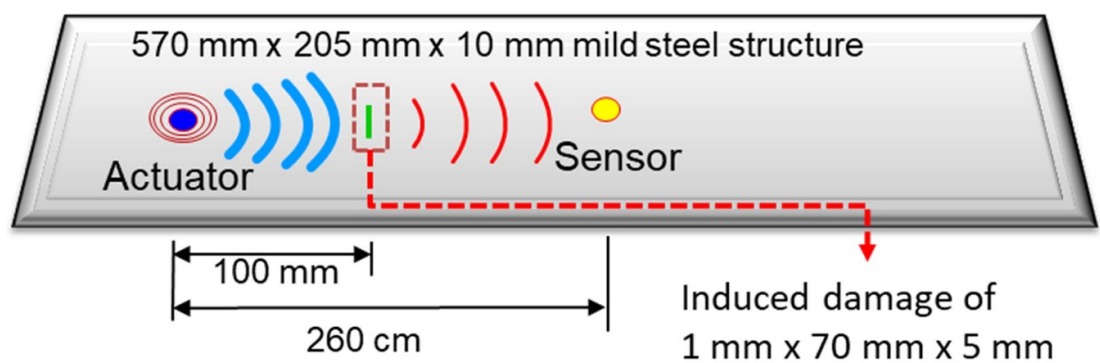


Figure 13. Induced damage in the structure.

The ESP32 published encrypted DI values over MQTTS under topic IUD_test/DI. Each message contained device identification, timestamp, and the computed DI. The HiveMQ dashboard successfully displayed encrypted packets and their decrypted equivalents in real time, as shown in figure 16, confirming correct operation of the AES-256 encryption, TLS transport, and broker-level synchronization.

For local visualization, a C#.NET desktop application subscribed to the same MQTT topic, retrieved the encrypted payloads, decrypted them using the shared AES-256 key and initialization vector, plotted the DI dynamically, and stored each decrypted message in a dedicated 'DataFolder' for post-analysis. The real-time visualization interface is shown in figure 17.

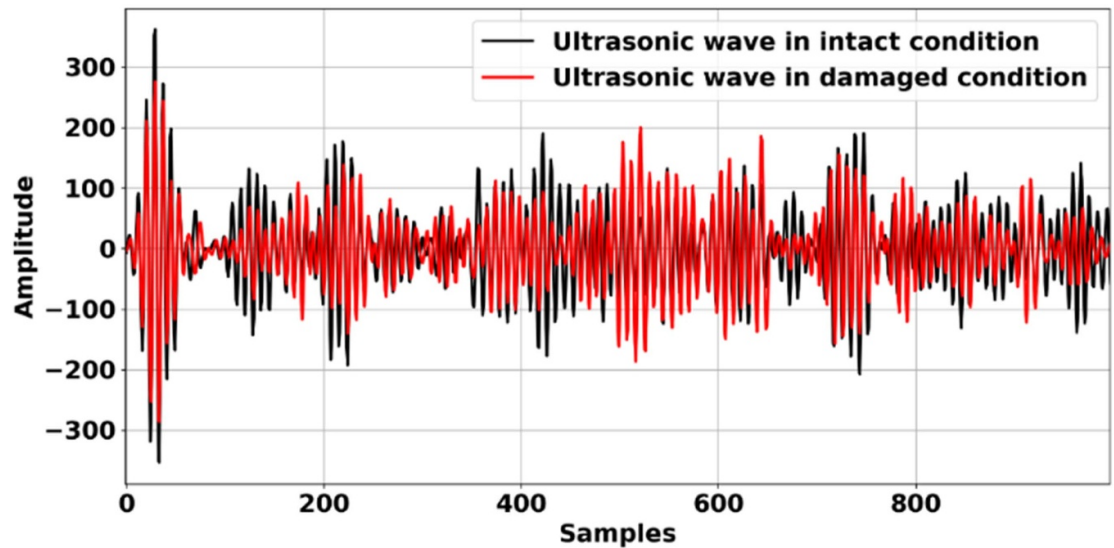


Figure 14. Comparison of the ultrasonic wave from the intact and damaged structure.

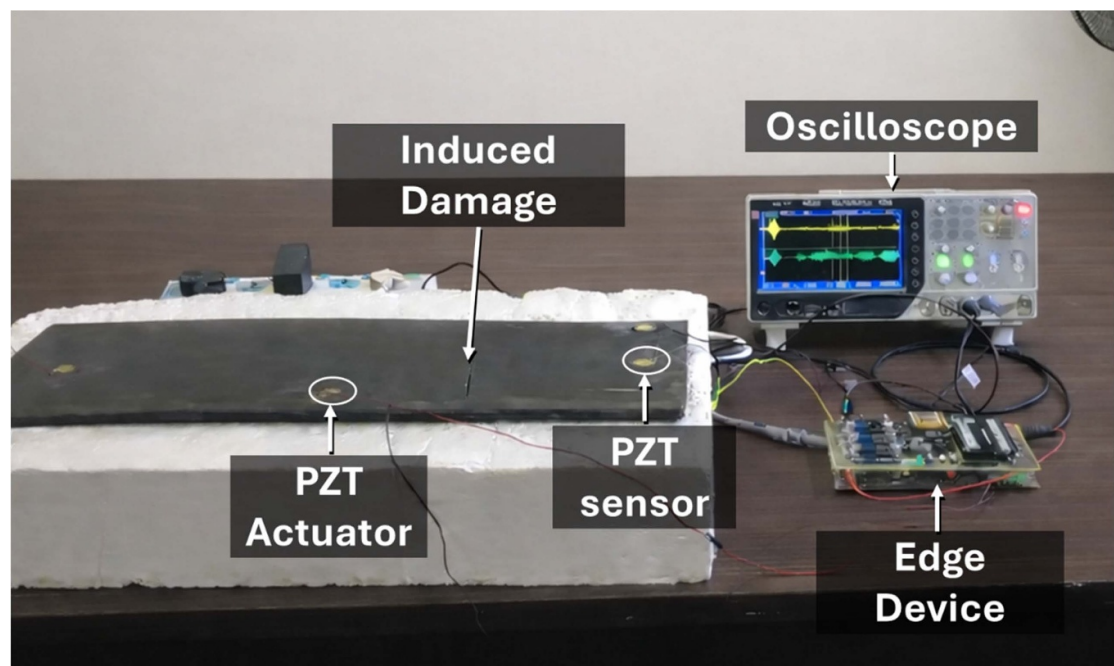


Figure 15. IoT setup for SHM.

Together, these cloud-based and desktop-based demonstrations confirm that the system supports secure, real-time monitoring and reliably transmits embedded DI computations without exposing raw ultrasonic waveforms.

3.4. Repeatability of DI measurements

To assess measurement repeatability, five trials were conducted for both intact and damaged conditions. DI values for the intact structure ranged from 0.78 to 0.92 (mean = 0.85, SD = 0.06), whereas DI values for the damaged structure ranged from 1.92 to 2.07 (mean = 2.01, SD = 0.06).

Table 3 summarizes these values, and the box plot in figure 18 shows a clear separation between the intact and damaged states with minimal overlap. The consistently higher DI in the damaged condition confirms the reliability and stability of the proposed SHM device for detecting and distinguishing structural health conditions.

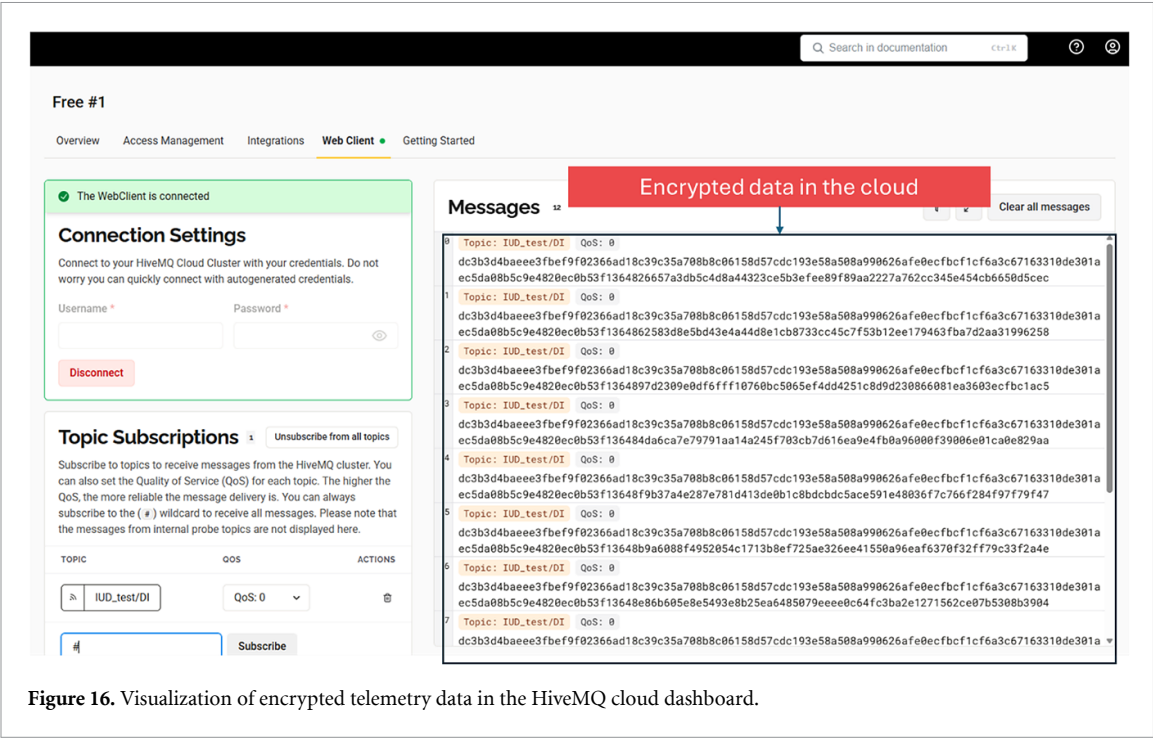


Figure 16. Visualization of encrypted telemetry data in the HiveMQ cloud dashboard.

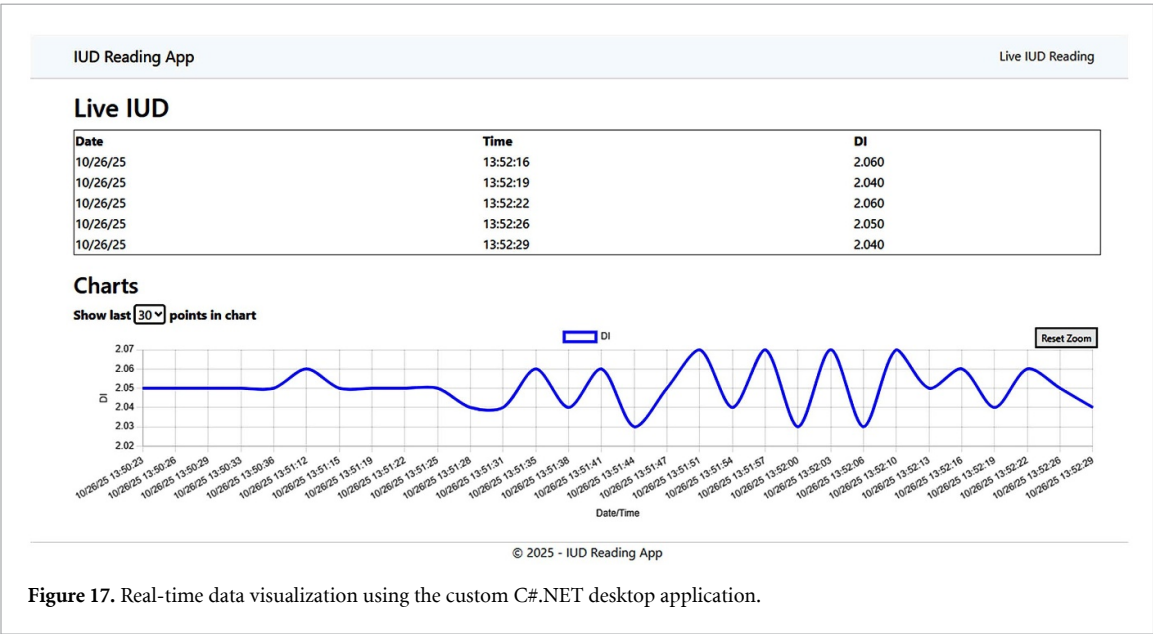


Figure 17. Real-time data visualization using the custom C#.NET desktop application.

Table 3. Damage index values for intact and damaged conditions.

Test number	Intact condition	Damaged condition
1	0.90	2.05
2	0.87	2.07
3	0.78	1.98
4	0.80	1.92
5	0.92	2.01

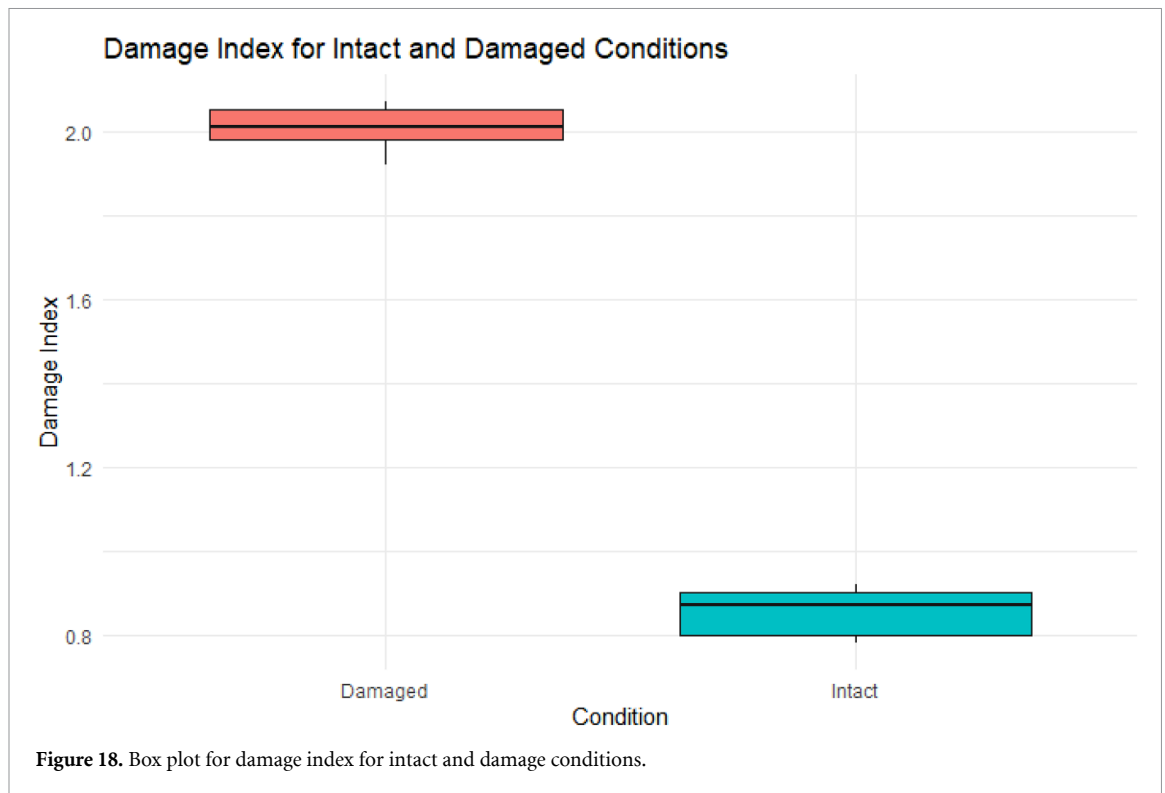


Table 4. Performance metrics of AES-256, TLS, and backoff algorithm.

Component	Parameter	Measured value	Observation
TLS	Handshake time	1,492,723 μ s	Efficient session establishment between device and MQTT broker.
	Heap usage (before $\rightarrow$ after)	283,228 $\rightarrow$ 243,776 bytes	$\Delta = -39\,452$ bytes, indicating released dynamic memory after secure connection setup.
AES-256 encryption	Execution time	656–664 μ s	Fast encryption of telemetry payload, suitable for real-time edge processing.
	Data length (before $\rightarrow$ after)	92 $\rightarrow$ 96 bytes	Increase due to PKCS7 padding.
	Ciphertext length	192 hex characters	Confirms proper 256-bit block encryption.
MQTT publish	API execution time	1,972–2,084 μ s	Stable and repeatable publishing time.
	Acknowledgment RTT	147,620–252,935 μ s	Consistent response latency under MQTTS session.
	Heap usage	243,452 bytes (no delta)	Memory remained stable during publish operation.
Exponential backoff	Latency increase	~ 300 μ s	Negligible impact while improving reconnection reliability.
Telemetry payload	Example	{'Org': 'IUD_STA1.0', 'Mdl': 'Hive', 'Ver': '1.1', 'Date': '10/26/25', 'Time': '13:51:12', 'DI': 2.06}	Typical message published under topic IUD_test/DI.

3.5. Processing overhead of AES-256, TLS, and the exponential backoff algorithm

The computational overhead associated with secure communication was analyzed in terms of execution time, memory usage, and transmission latency. Table 4 summarizes the performance.

- **TLS handshake:** approximately 1.49 s, with memory usage returning from 283 228 bytes to 243 776 bytes after session establishment.
- **AES-256 encryption:** 656–664 μ s per packet, with telemetry size increasing from 92 bytes to 96 bytes after PKCS7 padding.

- **MQTTs publish:** API execution time of 1972–2084 μs and acknowledgment round-trip times of 147–253 ms under MQTTS.
- **Exponential backoff:** added roughly 300 μs during reconnection events, a negligible impact relative to overall communication delays.

These metrics demonstrate that strong encryption and secure transport can be integrated into the edge device without compromising real-time sensing or telemetry responsiveness.

4. Conclusion

This work presented the design and implementation of a secure IoT-enabled edge computing device for ultrasonic SHM targeting inaccessible structural regions. The system integrates a PZT-based ultrasonic sensing mechanism with an STM32 microcontroller for on-board signal conditioning and DI computation, while an ESP32 enables secure wireless communication using AES-256, TLS (MQTTs), and an exponential backoff strategy. The system achieved an average encryption time of $\sim 660 \mu\text{s}$ and a TLS handshake time of $\sim 1.49 \text{ s}$, confirming the feasibility of secure, real-time operation. Encrypted telemetry was successfully transmitted to the HiveMQ cloud broker, and a custom C#.NET application was developed to subscribe, decrypt, and visualize the DI dynamically, providing both cloud-based and local real-time monitoring capabilities.

By performing all signal processing at the edge and transmitting only the computed DI, the proposed system reduces bandwidth requirements, latency, and exposure of raw structural data. Experimental validation demonstrated reliable detection of a $1 \times 70 \times 5 \text{ mm}$ crack in a mild steel specimen, confirming the effectiveness of the approach. Performance evaluation showed that secure, real-time operation is feasible, with minimal latency between acquisition and DI computation. This work demonstrates the effective integration of ultrasonic sensing, edge computing, and secure communication within a compact embedded platform.

The proposed system is suitable for continuous monitoring applications, as it minimizes communication overhead while maintaining real-time performance. With further enhancements in frequency range, multi-sensor integration, and advanced signal analysis techniques, the system can be extended to improve sensitivity to smaller defects and support scalable SHM deployments in complex infrastructures.

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Data availability statement

The datasets generated and analyzed during the current study are not publicly available due to size and internal storage limitations but are available from the corresponding author on reasonable request.

Conflict of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics statement

This research did not involve human participants, human data, human tissue, or animal experiments. Therefore, ethics approval and informed consent were not required.

Author contribution

Srijan Rajbamshi

Validation (supporting), Writing – review & editing (supporting)

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